

👉 XAI Lecture 20

Explainability for Fair ML & Robust Counterfactuals under the Right to be Forgotten

FAMAF - UNC

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eXplainable
Artificial Intelligence

Disclaimer -----

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

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<https://interpretable-ml-class.github.io/>

EXPLAINABILITY FOR FAIR MACHINE LEARNING

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ABSTRACT

As the decisions made or influenced by machine learning models increasingly impact our lives, it is crucial to detect, understand, and mitigate unfairness. But even simply determining what “unfairness” should mean in a given context is

(Begley et al. 2020)

The core tension in fair ML

- ML models make high-stakes decisions affecting individuals
- **Detecting unfairness is non-trivial:** many competing definitions exist
- Existing XAI tools **do not reliably indicate** whether a model is fair
- Using explainability to validate fairness can be misleading

Central Questions

- ① Can XAI tools detect unfairness?
- ② Can we **attribute** unfairness to individual features?
- ③ Can we use that attribution to **intervene** modularly?

🍷 Fairness Definitions: Examples -----

Demographic parity: $f(x)$ is unconditionally independent of sensitive attribute a

- If 100 female and 100 male students apply to Harvard, parity is achieved if the admission rate is equal for both groups, regardless of average qualification.

Equalized odds: $f(x)$ is independent of a given y

- Qualified female and male applicants have the same probability of admission; same for unqualified applicants.

Group A (Female)

	Qualified	Unqualified
Admitted	45	2
Rejected	45	8
Total	90	10



Group B (Male)

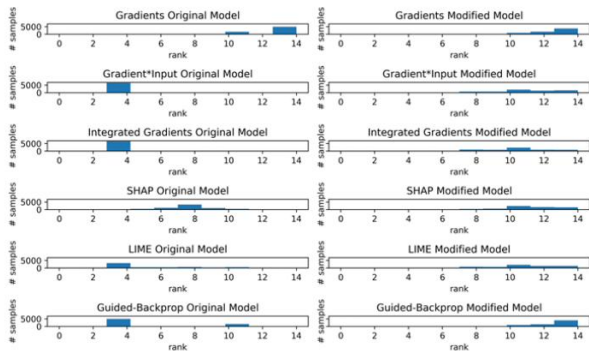
	Qualified	Unqualified
Admitted	5	18
Rejected	5	72
Total	10	90

🍌 The Explanation Attack -----

Explanation methods can be **manipulated**.

An explanation attack can easily mask a model's discriminatory use of a sensitive feature without hurting accuracy (Dimanov et al. 2020).

Importance ranking histograms for gender as the sensitive feature on the adult test set of original (left) and modified (right) models



Key Gap

If explanations can be manipulated, they cannot serve as evidence of fairness. Begley et al.'s solution: Fairness Shapley Values **must sum** to the chosen fairness metric — manipulation is impossible without changing the metric itself.

Background: Shapley Values -----

Cooperative game theory → feature attribution

$$\phi_v(i) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)]$$

Symbol	Meaning
$\phi_v(i)$	Shapley value of feature i
N	Set of all features
$S \subseteq N \setminus \{i\}$	Coalition not containing i

Symbol	Meaning
$v(S)$	Value of coalition S
$v(S \cup \{i\}) - v(S)$	Marginal contribution of i to S
$\frac{ S !(n- S -1)!}{n!}$	Weighting over all orderings

In Plain English

The Shapley value of feature i is its **average marginal contribution** when added to every possible coalition of other features — a fair division of the outcome among all participants.



Fair Machine Learning - Proposed Solution -----

- A **unified** approach that works for many **group-fairness** criteria
 - ▶ demographic parity, equalised odds, conditional demographic parity
 - ▶ for each definition, choose Shapley value functions that attribute overall fairness to individual features.
- **Cannot hide unfairness** by manipulating explanations
 - ▶ Fairness Shapley values collectively must sum to the chosen fairness metric

Explaining Model Accuracy -----

- Shapley value $\phi_v(i)$ attributes a portion to player i :

$$\phi_v(i) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)] \quad (1)$$

- Binary classification problem:

$$f_y(x) = (1 - y)(1 - f(x)) + y f(x) \quad (2)$$

- Value function by marginalising over out-of-coalition features:

$$v_{f_y(x)}(S) = \mathbb{E}_{p(x')} [f_y(x_S \cup x'_{N \setminus S})] \quad (3)$$

- Global explanation of model performance:

$$\Phi_f(i) = \mathbb{E}_{p(x,y)} [\phi_{f_y(x)}(i)] \quad (4)$$

🍷 Explaining Model Accuracy (cont) -----

- Aggregating global Shapley values:

$$\sum_i \Phi_f(i) = \underbrace{\mathbb{E}_{p(x,y)} [f_y(x)]}_{\text{Expected accuracy for a model which samples a predicted label according to the predicted probability}} - \underbrace{\mathbb{E}_{p(x')p(y)} [f_y(x')]}_{\text{The accuracy that is not attributable to any of the features and is related to the class balance}} \quad (5)$$

In Plain English

Total accuracy decomposes as a sum of individual feature contributions, with a baseline term capturing class imbalance.

Explaining Model Fairness -----

- To explain fairness, define a new value function that captures this effect :

Demographic parity calls for $f(x)$ to be unconditionally independent of a

$$g_a(x) = f(x) \cdot \frac{(-1)^a}{p(a)}, \quad a : \text{sensitive attribute} \quad (6)$$

- The value function on coalitions is defined through marginalisation:

$$v_{g_a(x)}(S) = \mathbb{E}_{p(x')} [g_a(x_S \cup x'_{N \setminus S})] \quad (7) \quad \Phi_g(i) = \mathbb{E}_{p(x,a)} [\phi_{g_a(x)}(i)] \quad (8)$$

- Each feature's marginal contribution to the overall demographic disparity:

$$\sum_i \Phi_g(i) = \int dx p(x|a=0) f(x) - \int dx p(x|a=1) f(x) \quad (9)$$

In Plain English

Each feature receives a share of the model's **total demographic disparity**. The sum of all Fairness Shapley Values equals exactly the chosen fairness metric — it cannot be manipulated without changing the metric.

Learning Corrective Perturbations -----

The **linearity axiom** of Shapley values guarantees that fairness Shapley values of a linear ensemble are the corresponding linear combination of the underlying models' values.

This motivates learning an additive perturbation to the original model:

$$f_{\theta} = f + \delta_{\theta}$$

$$\delta_{\theta}(f(x), x, a) = \sigma\left(\sigma^{-1}(f(x)) + \tilde{\delta}_{\theta}(f(x), x, a)\right) - f(x)$$

Symbol	Meaning
f	Original model (black box)
$\tilde{\delta}_{\theta}$	Any training-time fairness algorithm
f_{θ}	Corrected model

In Plain English

Instead of retraining the full model, a lightweight **patch** is learned to impose fairness. Shapley linearity ensures the corrected model's values remain interpretable in terms of its components.

Experimental Setup -----

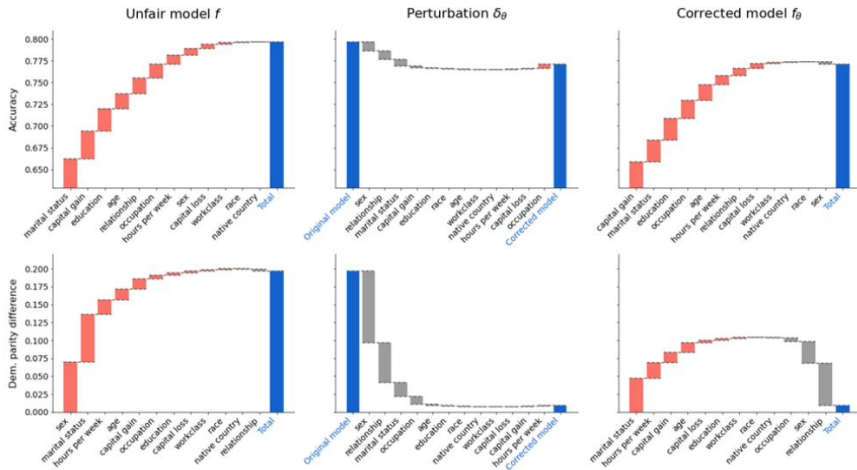
Datasets:

Dataset	Task	Sensitive attribute a
Adult (UCI)	Income prediction $\$ > \$50K$	Sex, Race
COMPAS	Recidivism prediction	Race

Evaluation: fairness metric and accuracy, before and after intervention.

🍷 Results: Explainability -----

On the Adult dataset, the features with the highest contribution to unfairness are **marital status**, **sex**, and **relationship** — even when **sex** does not appear directly in the model.

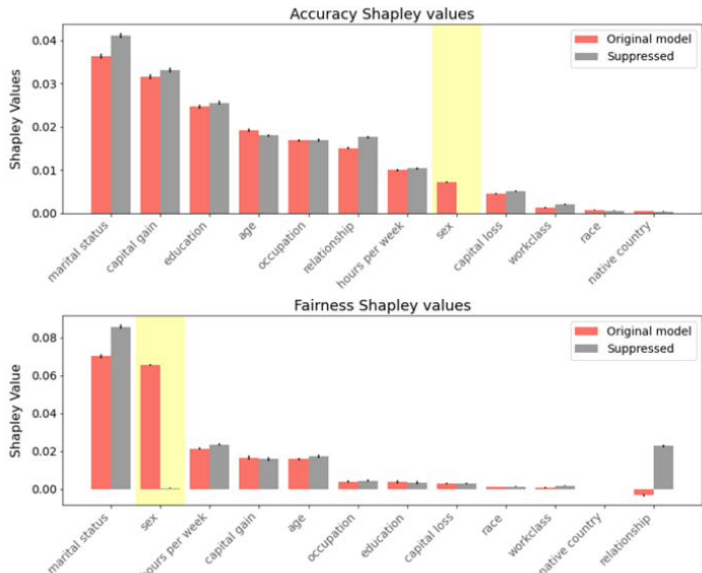




Results: Robustness of Fairness Explanations -----

An attack suppressing the importance of **sex** reduces the Demographic Parity Difference from 0.193 to 0.184 — a minimal reduction.

Fairness Shapley Values **cannot** be manipulated to hide unfairness without changing the global metric.



🍌 Results: Learned Perturbations (Demographic Parity) -----

Table 1: Accuracy associated with decreasing demographic parity thresholds.

		Accuracy [%] at demographic parity difference						
		0.1	0.08	0.06	0.04	0.02	0.01	0.005
Adult	Agarwal et al.	84.71	84.32	83.94	83.82	83.29	83.29	-
	Agarwal et al. - perturbed	84.69	84.43	83.82	83.82	83.35	83.23	-
	Zhang et al.	84.65	84.18	84.06	83.58	83.18	83.15	83.15
	Zhang et al. - perturbed	84.74	84.48	83.78	83.61	83.14	82.99	82.96
	Feldman et al. (post)	84.69	84.35	84.12	83.67	83.32	83.30	83.01
COMPAS	Agarwal et al.	74.05	74.05	73.77	73.67	73.11	73.11	73.01
	Agarwal et al. - perturbed	74.24	74.24	73.86	73.86	73.20	72.73	72.73
	Zhang et al.	75.19	75.19	75.19	74.62	74.15	74.15	74.15
	Zhang et al. - perturbed	74.24	74.24	74.24	73.30	73.30	73.20	72.73
	Feldman et al. (post)	74.81	74.81	74.81	74.24	74.24	73.20	72.35

- **No significant accuracy loss** across all thresholds
- Perturbed models track their originals closely
- Achieving near-perfect fairness costs less than **2 percentage** points of accuracy.

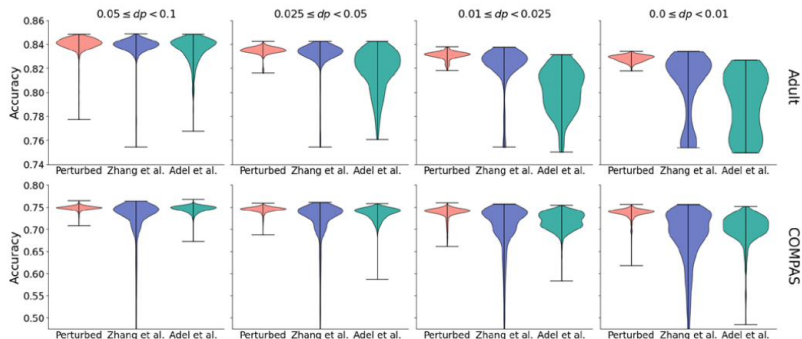
🍌 Results: Learned Perturbations (Equalized Odds) -----

Table 2: Accuracy associated with decreasing equalised odds thresholds.

		Accuracy [%] at equalised odds difference						
Method		0.1	0.08	0.06	0.04	0.02	0.01	0.005
Adult	Agarwal et al.	85.32	85.32	85.13	84.30	84.18	-	-
	Agarwal et al. - perturbed	85.43	85.43	85.31	84.34	84.21	-	-
	Zhang et al.	85.13	85.04	85.04	84.86	84.33	75.43	75.43
	Zhang et al. - perturbed	85.26	85.11	85.06	84.97	84.23	83.53	-
	Hardt et al.	82.77	82.77	82.77	82.77	82.77	82.77	82.77
COMPAS	Agarwal et al.	75.19	75.19	74.05	74.05	73.86	73.39	73.39
	Agarwal et al. - perturbed	74.43	74.43	74.43	73.86	73.39	73.39	73.39
	Zhang et al.	74.62	74.62	74.62	74.62	74.62	73.48	53.12
	Zhang et al. - perturbed	74.34	74.34	74.34	74.34	73.48	72.44	72.44
	Hardt et al.	71.31	71.31	71.31	70.45	68.75	-	-

- **No significant accuracy loss** under equalized odds
- Results generalize across fairness definitions

🍷 Results: Learned Perturbations -----



- No significant accuracy reduction under demographic parity or equalized odds
- The perturbative approach has **lower variance** and higher mean accuracy than baselines
- **Model-agnostic**: structure or access requirements apply only to the perturbation, not the original model

🍷 Limitations (Paper 1) -----

- The choice of disparity measure δ is **normative**, not technical
- Exact Shapley computation is **exponential** in $|F|$; approximations (SHAP) are used in practice
- The framework is **global**; local fairness attributions remain an open problem
- Does not resolve **which** fairness definition is appropriate for a given context

Notes

Does Shapley attribution *explain* unfairness, or merely *measure* it?

🍷 Summary of Contributions (Paper 1) -----

Contribution	Key point
Fairness Shapley Values	Attributes $\sum_i \Phi_g(i)$ to individual features
Linearity property	Unfairness decomposes additively
Meta-algorithm	Wraps any training-time fairness intervention
Robustness	Explanations cannot be manipulated

Towards Bridging the Gaps between the Right to Explanation and the Right to be Forgotten

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Abstract

The Right to Explanation and the Right to be Forgotten are two important principles outlined to regulate algorithmic decision making and data usage in real-world applications. While the right to explanation allows individuals to request an actionable explanation for an algorithmic decision, the right to be forgotten grants them the right to ask for their data to be deleted from all the databases and models of an organization. Intuitively, enforcing the right to be forgotten may trigger model updates which in turn invalidate previously provided explanations, thus violating the right to explanation. In this work, we investigate the technical implications arising due to the interference between the two aforementioned regulatory principles, and propose *the first algorithmic framework* to resolve the tension between them. To this end, we formulate a novel optimization problem to generate explanations that are robust to model updates due to the removal of training data instances by data deletion requests. We then derive an efficient approximation algorithm to handle the combinatorial complexity of this optimization problem.

(Krishna, Ma, and Lakkaraju 2023)

🍷 Motivation (Paper 2) -----

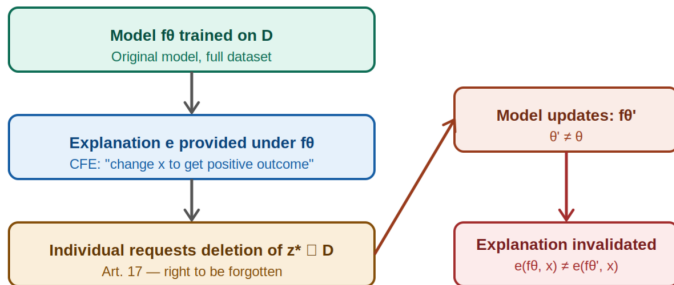
Two pillars of algorithmic accountability under GDPR

- **Art. 22 (Right to Explanation):**
individuals subject to automated decisions must receive a meaningful explanation
- **Art. 17 (Right to be Forgotten):**
individuals may request deletion of their personal data from models and databases
- Both rights are **legally enforceable simultaneously**

The Problem

- ① Both rights are legally binding
- ② Enforcing one may **automatically violate** the other
- ③ No existing framework addresses their **joint satisfaction**

🍵 The Conflict: A Concrete Example -----



- A bank trains f_{θ_1} on D ; denies a loan to x_0 ; provides counterfactual explanation x^* under f_{θ_1}
- Another user invokes Art. 17; their data is deleted; model updates to f_{θ_w}
- Explanation x^* is no longer valid under f_{θ_w} — **Art. 22 is violated**

🍷 Contributions (Paper 2) -----

- First algorithmic framework (**ROCERF**) to address the tradeoff between the right to explanation and the right to be forgotten
- ROCERF not only bridges the gap but also **outperforms** existing counterfactual explanation methods

RObust Counterfactual Explanations under the Right to be Forgotten

The framework proceeds in four steps:

- CFE as a standard optimization problem
- CFE as an optimization problem under data removal
- Efficient approximation (naive solution is intractable)
- Theoretical bounds on cost and validity

🍷 Notation: ROCERF -----

Data and classifier:

$$D = \{(x_i, y_i)\}_{i=1}^n, \quad x_i \in \mathcal{X}, \quad y_i \in \{-1, +1\}$$

$$f_{\theta_1} = \arg \min_{\theta} \frac{1}{n} \sum_{i=1}^n l_i(\theta), \quad f_{\theta_w} = \arg \min_{\theta} \frac{1}{\|w\|_1} \sum_{i=1}^n w_i l_i(\theta)$$

Data weight vector $w \in \{0, 1\}^n$:

$$w_i = \begin{cases} 1 & \text{data point } i \text{ in training set} \\ 0 & \text{data point } i \text{ deleted} \end{cases}$$

$w = \mathbf{1}$ means no data point has been removed.

Counterfactual Explanation (CFE) -----

- The Counterfactual Explanation is:
 - ▶ Optimisation problem - find valid counterfactual with minimum cost
 - ▶ the closest point to x_0 that flips the prediction:

$$\min_{x \in \mathcal{X}} \|x - x_0\|_2 \quad \text{s.t.} \quad f_{\hat{\theta}_1}(x) \geq 0$$

In Plain English

“What is the minimum change to your profile so the model approves your loan?” This counterfactual is valid for the current model — but may become invalid if the model changes.

🍷 k-Removal Robust CFE -----

Definition: a k-RR CFE is a counterfactual that remains valid upon removal of *any* k data points.

$$\mathcal{W}^{(k)} = \{w \in \{0, 1\}^n : \|w\|_1 = n - k\}$$

$$\min_{x \in \mathcal{X}} \|x - x_0\|_2 \quad \text{s.t.} \quad f_{\hat{\theta}_w}(x) \geq 0, \quad \forall w \in \mathcal{W}^{(k)}$$

Naive solution: train $\binom{n}{k}$ classifiers and optimise with $\binom{n}{k}$ constraints.

Key Gap

Computationally intractable. An efficient approximation is needed.

Efficient Approximation: First-Order Taylor -----

For fixed x , approximate $f_{\hat{\theta}_w}(x)$ via Taylor expansion w.r.t. w (Giordano et al., 2019):

$$\tilde{f}_{\hat{\theta}_w}(x) = f_{\hat{\theta}_1}(x) + \frac{1}{n} \sum_{i:w_i=0} \beta(x)^T H^{-1} g_i(\hat{\theta}_1)$$

where:

$$\beta(x) := \left(\frac{\partial f_{\theta}(x)}{\partial \theta} \Big|_{\theta=\hat{\theta}_1} \right)^T, \quad H := \frac{1}{n} \sum_{i=1}^n h_i(\hat{\theta}_1), \quad g_i(\theta) := \frac{\partial l_i(\theta)}{\partial \theta}$$

In Plain English

Instead of retraining for each possible deletion, the Hessian of the original model estimates how the prediction would change — in a single pass.

🍎 Eff. Approx: Reducing $\binom{n}{k}$ Constraints to One -----

The term $\beta(x)^T H^{-1} g_i(\hat{\theta}_1)$ is **independent of** w . Only the tightest constraint needs to be retained:

$$\mathcal{A}(x) := \{\beta(x)^T H^{-1} g_i(\hat{\theta}_1)\}_{i=1}^n$$

$$f_{\mathcal{A}}^{(k)}(x) := f_{\hat{\theta}_1}(x) + \frac{1}{n} \min_{\mathcal{B} \subseteq \mathcal{A}(x), |\mathcal{B}|=k} \sum_{b \in \mathcal{B}} b$$

$$\min_{x \in \mathcal{X}} \|x - x_0\|_2 \quad \text{s.t.} \quad f_{\mathcal{A}}^{(k)}(x) \geq \delta$$

In Plain English

The worst case is deleting the k points that most damage the prediction — pick the k smallest values of $\mathcal{A}(x)$.

🍷 Eff. Approx.: Final Optimisation Problem -----

- Solving the constrained optimization problem.
- Penalty method:

$$\phi(z) := \max(z, 0)^2$$

$$\min_{x \in \mathcal{X}} J_t(x) = \lambda_t \phi\left(\delta - f_{\mathcal{A}}^{(k)}(x)\right) + \|x - x_0\|_2$$

In Plain English

The constraint becomes a penalty term that grows until the solution satisfies $f_{\mathcal{A}}^{(k)}(x) \geq \delta$ — standard unconstrained optimisation.

Theoretical Guarantees: Linear Models -----

For regularised logistic regression with $l_i(\theta) = \log(1 + \exp(-y_i\theta^T x_i)) + \gamma\|\theta\|_2^2$:

$$\|\tilde{x}_0^{(k)} - x_0\|_2 \leq \|\tilde{x}_0 - x_0\|_2 + \frac{kC}{n\|\hat{\theta}_1\|_2}$$

The additional cost of robustness has an upper bound of $\mathcal{O}(k/n)$, with theoretical guarantees on validity.

In Plain English

Larger k (more deletions to tolerate) or smaller n (less data) increases the cost of the robust counterfactual — but the growth is controlled and bounded.

🍷 Theoretical Guarantees: Nonlinear Models -----

Under regularity assumptions (Lipschitz, convexity), for nonlinear models:

$$\|\tilde{x}_0^{(k)} - x_0\|_2 \leq \|\tilde{x}_0 - x_0\|_2 + \frac{2kC}{n}$$

If the function is μ -strongly convex, the bound improves further.

The linear approximation may not capture deep neural network behaviour — authors report cases where validity **improves** as more data is deleted, which is counterintuitive.

Experimental Setup -----

Datasets: three real-world binary classification datasets from high-stakes decision-making scenarios:

- **German Credit** (Dua & Graff, 2017): 1,000 individuals, 60 features (demographic, personal, financial). Target: credit risk — “good” or “bad”.
- **Adult** (Yeh & Lien, 2009): 48,842 individuals, features include demographics, education, employment, and financial data. Target: income — above or below \$50 k/year.
- **COMPAS** (Jordan & Freiburger, 2015): 18,876 defendants, criminal records and demographic features. Target: bail decision — “bail” or “no bail”.

Models:

- **Logistic Regression (LR):** regularised logistic regression (scikit-learn default). Accuracy: German Credit 72.2%, COMPAS 85.8%, Adult 84.0%.
- **Neural Network (NN):** 3-layer fully-connected feedforward network, hidden size = $2 \times$ input dim, centered-softplus activation, trained with SGD ($\text{lr} = 0.01$). Accuracy: German Credit 73.9%, COMPAS 85.1%, Adult 84.7%.

🍷 Experimental Setup cont. -----

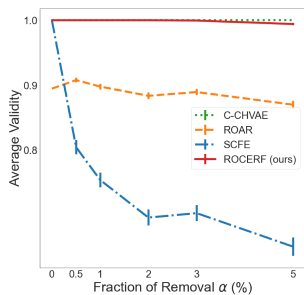
Baselines:

- **SCFE** (Wachter et al., 2017): gradient-based optimization to find the CFE closest to the input — solves the standard problem (1) without any robustness consideration.
- **C-CHVAE** (Pawelczyk et al., 2020): manifold-based method that searches for CFEs in a latent space, encouraging realistic counterfactuals on the data manifold.
- **ROAR** (Upadhyay et al., 2021): generates CFEs robust to small Gaussian perturbations of model parameters — the strongest baseline, as it addresses model changes but without guarantees under data deletion.

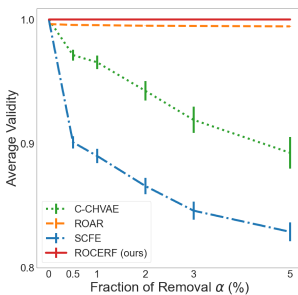
Protocol: randomly remove a fraction $\alpha \in [0.5\%, 5\%]$ of training data, repeated $M = 100$ times. Hyperparameters: $k = 0.5\%$ of training set size, $\delta = 0$.

🍷 Experimental Results: Logistic Regression -----

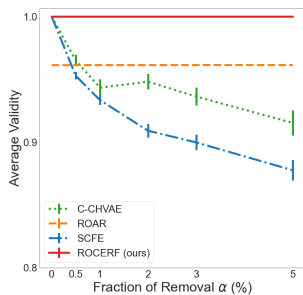
- The x-axis corresponds to the fraction of data removal α and the y-axis corresponds to the average validity.
- The error bars indicate the standard errors across $M = 100$ trials with each trial having an α fraction of training data points randomly removed.



(a) German Credit.



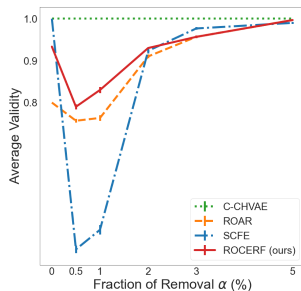
(b) COMPAS.



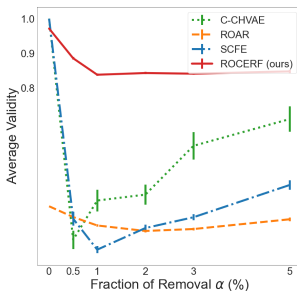
(c) Adult.

- ROCERF maintains validity ≈ 1.0 while all baselines degrade significantly.

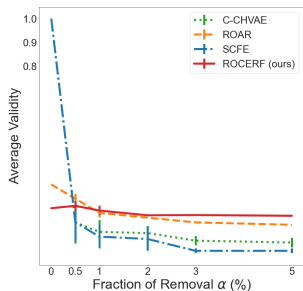
🍷 Experimental Results: Neural Networks Logistic Regression -----



(a) German Credit.



(b) COMPAS.



(c) Adult.

- More dataset-dependent behaviour.
 - ▶ German Credit shows counterintuitive improvement
 - ▶ Small dataset size causes dramatic boundary shifts after deletion.

🍷 Experimental Results: Average Cost -----

Methods	German Credit	COMPAS	Adult
SCFE	0.82 ± 0.12	0.78 ± 0.02	1.04 ± 0.006
C-CHVAE	8.51 ± 0.38	5.93 ± 0.11	3.79 ± 0.013
ROAR	1.45 ± 0.09	1.08 ± 0.01	1.07 ± 0.006
ROCERF (ours)	1.35 ± 0.14	0.87 ± 0.02	1.14 ± 0.006

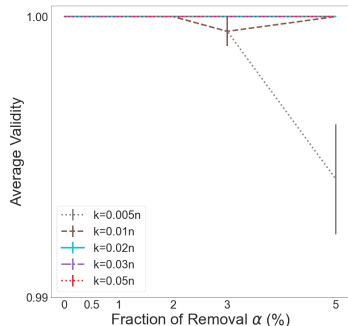
- **SCFE** has the lowest cost but significantly worse validity — cheap recourse that breaks under any deletion.
- **C-CHVAE** pays very high cost across all datasets — manifold constraint makes CFEs expensive.
- **ROAR** is close to **ROCERF** but **ROCERF** consistently achieves lower or equal cost with better validity.

🍷 Experimental Results: Average Cost Neural Networks -----

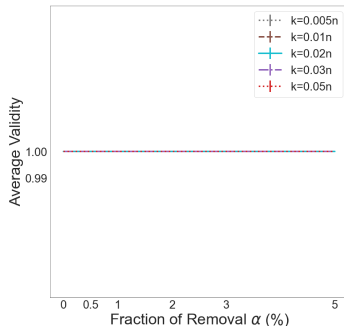
Methods	German Credit	COMPAS	Adult
SCFE	1.18 ± 0.08	0.97 ± 0.11	1.00 ± 0.09
C-CHVAE	4.45 ± 0.18	5.98 ± 0.12	8.83 ± 0.31
ROAR	3.84 ± 0.33	1.09 ± 0.13	4.07 ± 0.55
ROCERF (ours)	2.76 ± 0.22	3.07 ± 0.08	4.06 ± 0.52

- **C-CHVAE** cost explodes on Adult (8.83) — manifold search fails badly in complex model settings.
- **ROAR** and **ROCERF** track closely on COMPAS and Adult — consistent with the validity results.
- **ROCERF** achieves the best cost/validity tradeoff overall: robustness does not require paying a large extra cost.

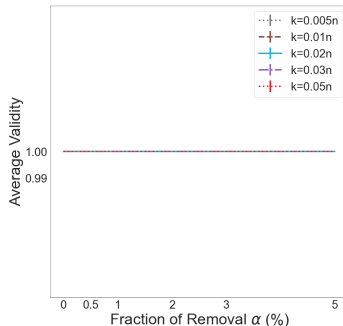
🍲 Sensitivity Analysis - Hyperparameter k -----



(a) German Credit.



(b) COMPAS.



(c) Adult.

- **COMPAS and Adult:** all variants of k achieve 100% validity across all α values.
- **German Credit:** lowest k shows a slight drop for high α — fixed by increasing k .
- Key guarantee: $k = 0.01n$ achieves 100% validity for any $\alpha \leq 1\%$; $k = 0.02n$ for any $\alpha \leq 2\%$.

🍲 The Validity-Cost Trade-off -----

Forcing an explanation to remain valid under **more possible deletions** necessarily moves the counterfactual further from the original point. This is not a bug — it is the **correct price** for regulatory compliance. The cost grows as k increases.

For Regulators

How much additional cost is acceptable in exchange for deletion-robust explanations? This is a **policy decision**, not a technical one.

🍷 Limitations (Paper 2) -----

- Taylor approximation is exact only for convex losses; nonlinear models exhibit counterintuitive behaviour
- Inverting H is $\mathcal{O}(p^3)$ — impractical for large neural networks
- k must be specified in advance; no principled selection method is provided
- ROAR (with the same hyperparameters) tracks ROCERF closely — is ROCERF really necessary?

Philosophical Note

An explanation that changes when training data changes was never really about the *individual* — it was about the *model state at a given moment*.

🍷 Summary of Contributions (Paper 2) -----

Contribution	Key point
ROCERF framework	First algorithm bridging the Art. 22 / Art. 17 conflict
k-RR CFE	Counterfactual valid under any deletion of k data points
$\mathcal{O}(n)$ approximation	Taylor + infinitesimal jackknife reduces $\binom{n}{k}$ constraints to one
Theoretical guarantees	Additional cost bounded by $\mathcal{O}(k/n)$ for linear models

🍷 Thank You! -----

Thank You!

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